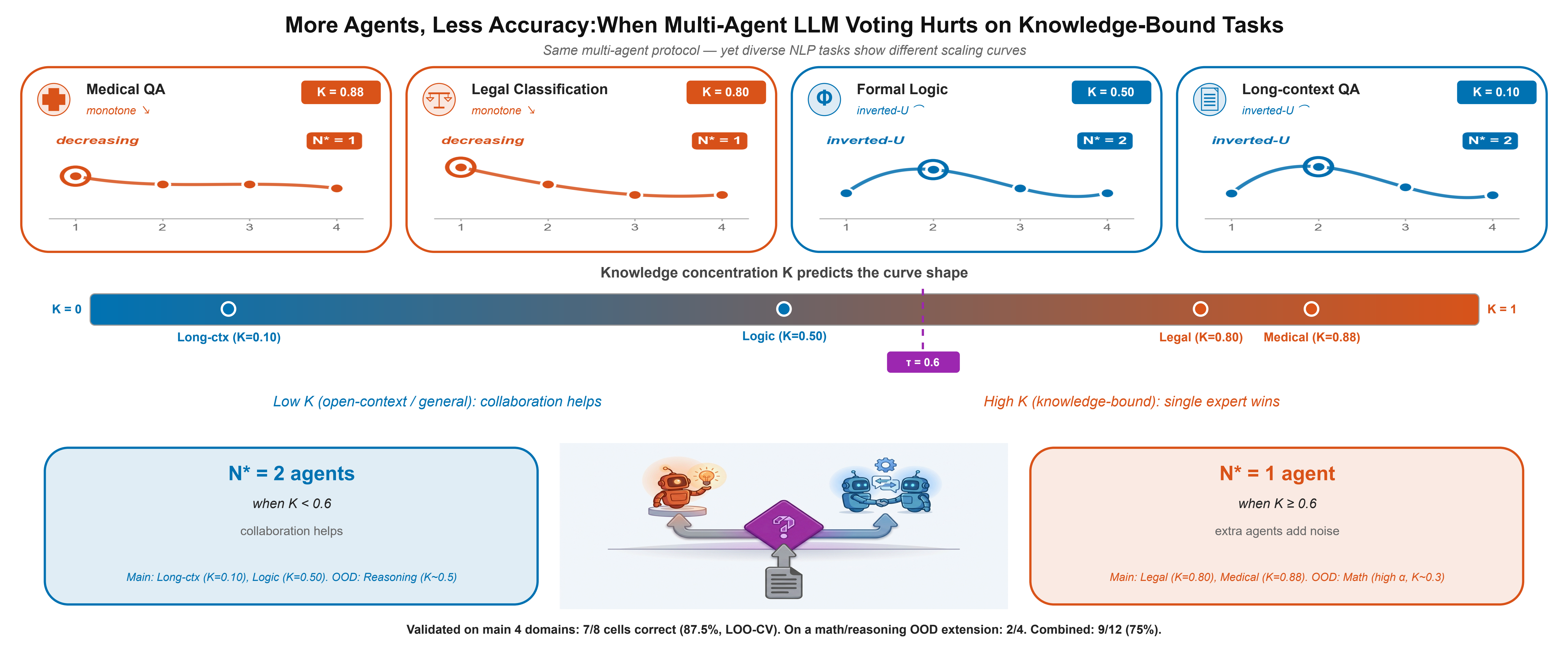

More Agents, Less Accuracy:When Multi-Agent LLM Voting Hurts on Knowledge-Bound Tasks
Same multi-agent protocol — yet diverse NLP tasks show different scaling curves
Medical QA
monotone ↘
K = 0.88
decreasing
N* = 1
1
2
3
4
Legal Classification
monotone ↘
K = 0.80
decreasing
N* = 1
1
2
3
4
Φ
Formal Logic
inverted-U ⌒
K = 0.50
inverted-U
N* = 2
1
2
3
4
Long-context QA
inverted-U ⌒
K = 0.10
inverted-U
N* = 2
1
2
3
4
Knowledge concentration K predicts the curve shape
K = 0
K = 1
Long-ctx (K=0.10)
Logic (K=0.50)
Legal (K=0.80)
Medical (K=0.88)
τ = 0.6
Low K (open-context / general): collaboration helps
High K (knowledge-bound): single expert wins
N* = 2 agents
when K < 0.6
collaboration helps
Main: Long-ctx (K=0.10), Logic (K=0.50). OOD: Reasoning (K~0.5)
N* = 1 agent
when K ≥ 0.6
extra agents add noise
Main: Legal (K=0.80), Medical (K=0.88). OOD: Math (high α, K~0.3)
Validated on main 4 domains: 7/8 cells correct (87.5%, LOO-CV). On a math/reasoning OOD extension: 2/4. Combined: 9/12 (75%).